

Multiple Choice (10 marks)

1. __ refers to a model that can neither model the training data nor generalize to new data.
 - (a) good fitting
 - (b) overfitting
 - (c) underfitting
 - (d) all of the above
2. Which of the following is NOT supervised learning?
 - (a) PCA
 - (b) Decision Tree
 - (c) Linear Regression
 - (d) Naive Bayesian
3. Which image data augmentation is most common for natural images?
 - (a) random crop and horizontal flip
 - (b) random crop and vertical flip
 - (c) posterization
 - (d) dithering
4. Predicting the amount of rainfall in a region based on various cues is a _____ problem.
 - (a) Supervised learning
 - (b) Unsupervised learning
 - (c) Clustering
 - (d) None of the above
5. Which of the following tasks can be best solved using Clustering.
 - (a) Predicting the amount of rainfall based on various cues
 - (b) Detecting fraudulent credit card transactions
 - (c) Training a robot to solve a maze
 - (d) All of the above

True / False (10 marks)

Answer True or False

6. True or False: DCGANs employ self-attention mechanisms to enhance the quality of generated images during training.
7. True or False: The original ResNet employs self-attention mechanisms similar to those found in Transformers.
8. True or False: Using an L 2 norm regularization penalty in linear regression can result in some coefficients being zeroed out.
9. True or False: The joint probability $P(H, U, P, W)$ can be expressed as $P(H) * P(W) * P(P | W) * P(U | H, P)$.
10. True or False: The classification run time of the nearest neighbors algorithm is $O(N)$, where N is the number of instances in the training dataset.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss how the choice of polynomial degree in polynomial regression influences the balance between underfitting and overfitting, providing examples to support your analysis.
12. Discuss the importance of centering the data to zero mean in Principal Component Analysis (PCA) and explain how this step enables PCA to yield projections equivalent to those obtained via Singular Value Decomposition (SVD).

End of Paper